

Business Problem

A retail company is experiencing noticeable shifts in customer purchasing behavior across demographics, product categories, and sales channels (online vs. offline). Management wants to better understand: What drives purchase decisions? What influences repeat purchases? How discounts, reviews, seasonality, and payment methods impact sales? Which customer segments generate the most value ?

Core Business Question

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

The Objective of this project is to transform raw customer data into actionable insights that support data-driven decision making across marketing, sales,, and product strategy.

Analytical Approach & Deliverables

◆ Data Preparation & Modeling (Python)

Cleaned and transformed raw transactional data using Python, handling missing values and inconsistencies. Engineered relevant features and structured the dataset for SQL analysis and BI reporting.

◆ Data Analysis (SQL)

Organized the dataset into a structured format and executed SQL queries to analyze customer segments, repeat purchases, product performance, discount impact, and subscription trends. Extracted measurable insights to support business decisions.

◆ Visualization & Dashboard (Power BI)

Built an interactive Power BI dashboard to visualize KPIs, revenue trends, customer demographics, and online vs. offline performance. Designed for stakeholder-friendly, data-driven exploration.

◆ Insights, Recommendations & Presentation

Synthesized findings into actionable business recommendations to improve engagement, retention, and revenue. Delivered both a written report and a stakeholder presentation to clearly communicate insights and strategic next steps.

Project Summary (TL;DR)

Analyzed retail consumer behavior data using Python, SQL, and Power BI to identify purchase drivers, segment customers, and evaluate sales channel performance. Delivered actionable insights and strategic recommendations supported by an interactive dashboard and stakeholder presentation.